Euler's Method and Logistic Growth

ANSWER KEY

Section 1: Euler's Method Iterations

1 Given $\frac{dy}{dx} = x + y$ and initial condition $y(0) = 1$, use Euler's method with step size $\Delta x = 0.5$ to approximate $y(1)$.

a. 2.25 ✓

b. 2.00

c. 2.50

d. 2.75

Step 1: at $(0, 1)$, slope $= 0 + 1 = 1$. $\Delta y = 1(0.5) = 0.5 \Rightarrow y(0.5) \approx 1.5$. Step 2: at $(0.5, 1.5)$, slope $= 0.5 + 1.5 = 2$. $\Delta y = 2(0.5) = 1.0 \Rightarrow y(1) \approx 1.5 + 1.0 = 2.25$.

2 If $y'' > 0$ (concave up) on the interval, does Euler's method yield an overestimate or underestimate of the true solution?

a. **An underestimate because tangent segments lie below the curve** ✓

b. An overestimate because tangent segments lie above the curve

c. Exact value

d. Depends on the step size

Euler's method uses tangent line segments. When the solution curve is concave up ($y'' > 0$), the curve bends above the tangent segments, making Euler's method an underestimate.

3 Given $\frac{dy}{dx} = 2x$ with $y(1) = 3$, what is the error in approximating $y(2)$ using Euler's method with $\Delta x = 0.5$?

a. 0.5 ✓

b. 1.0

c. 0

d. 0.25

Euler: $y(1.5) = 3 + 2(1)(0.5) = 4$. $y(2) = 4 + 2(1.5)(0.5) = 4 + 1.5 = 5.5$. Exact: $y = x^2 + 2 \Rightarrow y(2) = 6$. Error $= 6 - 5.5 = 0.5$.

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Section 2: The Logistic Differential Equation

4 A population grows according to $\frac{dP}{dt} = 0.04P(1 - \frac{P}{500})$. What is the carrying capacity of this population?

a. 500 ✓

b. 0.04

c. 250

d. $12,500$

In the standard logistic equation $\frac{dP}{dt} = kP(1 - \frac{P}{M})$, the parameter M is the carrying capacity, which is 500.

5 For the logistic model $\frac{dP}{dt} = 0.04P(1 - \frac{P}{500})$, at what population size is the growth rate $\frac{dP}{dt}$ maximized?

a. 250 ✓

b. 500

c. 125

d. 375

The rate of growth is maximized at the point of inflection of the logistic curve, which occurs at exactly half the carrying capacity: $\frac{M}{2} = \frac{500}{2} = 250$.

6 What is $\lim_{t \to \infty} P(t)$ for a logistic population with carrying capacity 800 if $P(0) = 50$?

a. 800 ✓

b. ∞

c. 400

d. 50

For any positive initial population $P_0 > 0$, the logistic curve asymptotically approaches the carrying capacity $M = 800$ as $t \to \infty$.

Section 3: Analytical Solutions to Logistic Models

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7 What is the general form of the solution to the logistic differential equation $\frac{dP}{dt} = kP(1 - \frac{P}{M})$?

a. $P(t) = \frac{M}{1 + Ae^{-kt}}$ ✓

b. $P(t) = Me^{kt}$

c. $P(t) = M - Ae^{-kt}$

d. $P(t) = \frac{M}{1 - Ae^{kt}}$

By separation of variables and partial fractions, the logistic solution is $P(t) = \frac{M}{1 + Ae^{-kt}}$.

8 If $P(t) = \frac{1000}{1 + 9e^{-0.2t}}$, what is the initial population $P(0)$?

a. 100 ✓

b. 1000

c. 90

d. 10

$P(0) = \frac{1000}{1 + 9(1)} = \frac{1000}{10} = 100$.

9 What is the concavity of the logistic curve when $P(t) > \frac{M}{2}$?

a. **Concave down ($P''(t) < 0$)** ✓

b. Concave up ($P''(t) > 0$)

c. Zero concavity

d. Undefined

For $P < M/2$, the curve is concave up (accelerating growth). For $P > M/2$, the curve is concave down (decelerating growth).

Section 4: Advanced Slope Fields with Autonomous Equations

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10 In the autonomous differential equation $\frac{dy}{dx} = y(y-3)(y+2)$, what are all equilibrium solutions?

a. $y = 0, y = 3, y = -2$ ✓

b. $y = 3, y = -2$ only

c. $y = 0$ only

d. $x = 0, x = 3, x = -2$

Equilibrium solutions occur where $\frac{dy}{dx} = 0$, giving horizontal lines $y = 0, y = 3,$ and $y = -2$.

11 For $\frac{dy}{dx} = 2y(1 - \frac{y}{4})$, which equilibrium solution is stable (an attractor)?

a. $y = 4$ ✓

b. $y = 0$

c. Both are stable

d. Neither is stable

Solutions starting near $y = 4$ approach 4 as $x \to \infty$ (since $y' > 0$ for $0 < y < 4$ and $y' < 0$ for $y > 4$), making $y = 4$ a stable attractor.

12 If $y(0) = 6$ in $\frac{dy}{dx} = 2y(1 - \frac{y}{4})$, what is $\lim_{x \to \infty} y(x)$?

a. 4 ✓

b. 6

c. ∞

d. 0

Since $y(0) = 6 > 4$, $y' < 0$, so $y(x)$ decreases asymptotically toward the stable carrying capacity 4.

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